Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Pandas Library : Functions

Date: 08-01-2026

Pandas

Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language

Series

A one-dimensional labeled array holding data of any type such as integers, strings, Python objects etc.

DataFrame

A two-dimensional data structure that holds data like a two-dimension array or a table with rows and columns.

Table 1: Commonly Used Pandas Functions and Their Descriptions

Function	Description
pd.read_csv()	Reads a CSV file into a Pandas DataFrame.
DataFrame.head()	Displays the first few records of a DataFrame.
DataFrame.tail()	Displays the last few records of a DataFrame.
DataFrame.shape	Returns the number of rows and columns in the DataFrame.
DataFrame.info()	Provides a concise summary of the DataFrame.
DataFrame.describe()	Generates descriptive statistics of numerical columns.
DataFrame.isnull()	Detects missing values in the DataFrame.
DataFrame.fillna()	Fills missing values with a specified value.
DataFrame.dropna()	Removes rows or columns containing missing values.
DataFrame.groupby()	Groups data based on one or more attributes.
DataFrame.mean()	Computes the mean of numerical columns.
DataFrame.sum()	Computes the sum of numerical columns.
DataFrame.min()	Returns the minimum value in each column.
DataFrame.max()	Returns the maximum value in each column.
DataFrame.sort_values()	Sorts the DataFrame based on specified column(s).
DataFrame.loc[]	Accesses rows and columns by labels.
DataFrame.iloc[]	Accesses rows and columns by integer positions.

Function	Description
DataFrame.drop()	Deletes specified rows or columns.
DataFrame.rename()	Renames column or index labels.
DataFrame.apply()	Applies a function along an axis of the DataFrame.
DataFrame.value_counts()	Counts unique values in a column.
DataFrame.nunique()	Returns the number of unique values in each column.
DataFrame.astype()	Converts data types of columns.
DataFrame.to_csv()	Writes the DataFrame to a CSV file.

For details on operations in pandas, see the official documentation: `https://pandas.pydata.org/docs/user_guide/10min.html#`